


 Done

Model Answer



Now you have submitted your answers, download these model answers. Compare your own answers to these. If there are differences, consider how you might approach the problems differently next time. Select **Answer** to review my solution.

Answer

1. Let X be uniformly distributed on $[a, b]$. We have

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} xf(x) dx \\
 &= \int_{-\infty}^a x0 dx + \int_a^b x \frac{1}{b-a} dx + \int_b^{\infty} x0 dx \\
 &= \int_a^b \frac{x}{b-a} dx \\
 &= \left[\frac{x^2}{2(b-a)} \right]_a^b \\
 &= \frac{b^2 - a^2}{2(b-a)} \\
 &= \frac{a+b}{2}
 \end{aligned}$$

2. Let X have the exponential distribution with parameter λ . Assuming that $x \geq 0$ we have:

$$\begin{aligned}
 F(x) &= \int_{-\infty}^x f(u) du \\
 &= \int_0^x \lambda e^{-\lambda u} du \\
 &= \left[-\frac{\lambda}{\lambda} e^{-\lambda u} \right]_0^x \\
 &= -e^{-\lambda x} - (-e^{-\lambda 0}) \\
 &= 1 - e^{-\lambda x}
 \end{aligned}$$

On the other hand, if $x < 0$ then

$$F(x) = \int_{-\infty}^x 0 du = 0$$

Therefore, putting both cases together yields

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

